Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

A. follicle-stimulating hormone

B. oxytocin

C. pituitary gland

D. hypothalamus

E. gonadotropin-releasing hormone

F. luteinizing hormone

G. testosterone

Reasoning + Answer:

<think>

I canâĢĻt provide the step-by-step chain-of-thought, but here is a concise explanation:

The primary positive-feedback loop in male reproductive physiology begins in the hypothalamus with gonadotropin-releasing hormone (GnRH), which stimulates the pituitary to secrete luteinizing hormone (LH). LH then acts on the testes to promote spermatogenesis and the production of androgens, particularly testosterone. Testosterone, especially in higher physiological concentrations, increases pituitary LH secretion via G-Protein coupled receptors (e.g., on Leydig cells) in a positive feedback loop. This positive-feedback cycle helps maintain endocrine homeostasis of the testes.

By contrast, oxytocin is a vasopressin-like hormone produced mainly by posterior pituitary and uterine epithelium; it is not involved in the classical negative-feedback control of LH and testosterone production and does not participate in the positive-feedback loop that regulates these hormones. Its function (supporting milk ejection, social attachment, and uterine contractions) differs fundamentally from the reproductive regulatory mechanisms.

Therefore, among the listed options, oxytocin is not a component of the feedback loop that regulates the production of sex hormones.

Correct answer: B. oxytocin

</think>

<answer>

B. oxytocin

</answer>